STAT GU4204 Midterm Cheat Sheet

Key Definitions

Statistical inference: Procedure producing a probabilistic statement about parts of a statistical model.

Parameter: Characteristic(s) determining the joint distribution of the r.v.'s of interest.

Parameter space Ω: Set of all possible values of θ.

Statistic: $T = \varphi(X_1, \ldots, X_n)$ — function of data ONLY, no unknown parameters.

Estimator: $\hat{\theta}_n = \varphi(X_1, \ldots, X_n)$ — a statistic used to estimate a parameter; it is a *random variable*.

Estimate: $\varphi(x_1, \ldots, x_n)$ — the realized *number* after plugging in observed data.

Bias: $b(T_n) = E(T_n) - g(\theta)$.

Unbiased: $E(T_n) = g(\theta)$ for ALL $\theta \in \Omega$ (bias $= 0$ everywhere).

Consistent: $\hat{\theta}_n \xrightarrow{P} \theta$ as $n \to \infty$. Large-sample property; does NOT tell us about finite-n performance.

MSE: $E[(T_n - g(\theta))^2] = \mathrm{Var}(T_n) + [\mathrm{bias}(T_n)]^2$.

Dominates: T_n dominates S_n if $\mathrm{MSE}(T_n) \le \mathrm{MSE}(S_n)$ for ALL θ.

Inadmissible: S_n is inadmissible if some T_n dominates it.

MVUE: Minimum Variance Unbiased Estimator — unbiased AND has smallest variance among all unbiased estimators.

Efficient estimator: Unbiased estimator that achieves the Cramér-Rao lower bound.

Sufficient statistic: T is sufficient for θ if the conditional distribution of $X|T{=}t$ does NOT depend on θ.

Likelihood: $L(\theta) = \prod f(X_i, \theta)$.

Log-likelihood: $\ell(\theta) = \sum \log f(X_i, \theta)$.

Score function: $\dot{\ell}(x, \theta) = \frac{\partial}{\partial\theta} \log f(x, \theta)$.

Score equation: $\sum \dot{\ell}(X_i, \theta) = 0$ (solve for MLE).

Fisher information: $I(\theta) = E[\dot{\ell}^2(X, \theta)] = \mathrm{Var}[\dot{\ell}(X, \theta)]$. Score has mean 0.

Sampling distribution: Distribution of a statistic $T = \varphi(X_1, \ldots, X_n)$.

Pivot: $\Psi(\mathbf{X}_n, g(\theta))$ — function of data AND parameter whose distribution does NOT depend on θ.

Confidence interval: Random interval (A, B) such that $P_\theta(A \le g(\theta) \le B) \ge 1 - \alpha$ for all θ. **Exact** if equality holds.

Prior $\xi(\theta)$: Belief about θ BEFORE observing data.

Posterior $\xi(\theta|x)$: Updated belief about θ AFTER observing data.

Conjugate prior: Prior such that posterior belongs to the SAME family as the prior.

Bayes estimator: Minimizes expected posterior loss.

- Squared error loss $\Rightarrow$ posterior mean
- Absolute error loss $\Rightarrow$ posterior median

Distributions Reference

Dist	PDF/PMF	$E(X)$	$\mathrm{Var}(X)$	MGF
$\mathrm{Ber}(p)$	$p^x(1-p)^{1-x}$	p	$p(1-p)$	$1-p+pe^t$
$\mathrm{Bin}(n,p)$	$\binom{n}{k}p^k(1-p)^{n-k}$	np	$np(1-p)$	$(1-p+pe^t)^n$
$\mathrm{Poi}(\lambda)$	$e^{-\lambda}\lambda^k/k!$	λ	λ	$e^{\lambda(e^t-1)}$
$\mathrm{Exp}(\theta)$	$\theta e^{-\theta x}$	$1/\theta$	$1/\theta^2$	$\theta/(\theta-t)$
$N(\mu,\sigma^2)$	$\frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2	$e^{\mu t+\sigma^2t^2/2}$
$\mathrm{Unif}[a,b]$	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{tb}-e^{ta}}{t(b-a)}$
$\mathrm{Gam}(\alpha,\beta)$	$\frac{\beta^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\beta x}$	α/β	α/β^2	$\left(\frac{\beta}{\beta-t}\right)^\alpha$
χ^2_d	$= \mathrm{Gam}(d/2, 1/2)$	d	$2d$	$(1-2t)^{-d/2}$
t_n	symm, heavy tails	0	$\frac{n}{n-2}$	—
$\mathrm{Beta}(\alpha,\beta)$	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	—

Gamma Function

$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1}e^{-x}\,dx$

$\Gamma(\alpha{+}1) = \alpha\,\Gamma(\alpha)$ $\quad$ $\Gamma(n) = (n{-}1)!$ $\quad$ $\Gamma(1/2) = \sqrt{\pi}$

Reproductive Properties (Indep. Sums)

X_i iid from	$\sum X_i \sim$
$\mathrm{Ber}(p)$	$\mathrm{Bin}(n,p)$
$\mathrm{Bin}(n_i,p)$	$\mathrm{Bin}(\sum n_i, p)$
$\mathrm{Gam}(\alpha_i,\beta)$ same β	$\mathrm{Gam}(\sum\alpha_i,\beta)$
$\chi^2_{m_i}$	$\chi^2_{\sum m_i}$
$\mathrm{Poi}(\lambda_i)$	$\mathrm{Poi}(\sum\lambda_i)$
$N(\mu_i,\sigma_i^2)$	$N(\sum\mu_i, \sum\sigma_i^2)$

Normal Properties

$X \sim N(\mu,\sigma^2) \Rightarrow aX + b \sim N(a\mu + b,\ a^2\sigma^2)$

$\bar{X}_n \sim N(\mu, \sigma^2/n)$ for iid normal sample.

CDF: $\Phi(z)$, $\quad$ symmetry: $1 - \Phi(x) = \Phi(-x)$

Key values: $z_{0.025} = 1.96$, $\quad z_{0.05} = 1.645$, $\quad z_{0.005} = 2.576$

Probability Inequalities

Markov: $W \ge 0 \;\Rightarrow\; P(W \ge t) \le E(W)/t$

Chebyshev: $P(|Z - EZ| \ge \tau) \le \mathrm{Var}(Z)/\tau^2$

On $\bar{X}$: $P(|\bar{X} - \mu| \ge \tau) \le \sigma^2/(n\tau^2)$

Sample size: want $P(|\bar{X} - \mu| \ge \varepsilon) \le \delta$, need $n \ge \sigma^2/(\varepsilon^2\delta)$.

Convergence

In probability ($Z_n \xrightarrow{P} Z$): $P(|Z_n - Z| > \varepsilon) \to 0$ for all $\varepsilon > 0$.

In distribution ($Z_n \xrightarrow{d} Z$): $F_n(u) \to F(u)$ at all continuity points of F.

Conv in prob $\Rightarrow$ conv in dist. Converse is FALSE in general (TRUE if limit is a constant).

WLLN: X_i iid, $E|X_1| < \infty \;\Rightarrow\; \bar{X}_n \xrightarrow{P} \mu$.

Conditions: iid, finite mean. Proof uses Chebyshev (if variance exists).

SLLN: $P(\lim \bar{X}_n = \mu) = 1$ $\quad$ (almost sure convergence, STRONGER).

CLT: X_i iid, mean μ, $\sigma^2 < \infty$:

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0,1)$$

Conditions: iid, finite variance. Gives APPROXIMATE dist for large n. Does NOT require normality of X_i.

Continuous Mapping Theorem:

- $Z_n \xrightarrow{P} b$, g cts at $b \Rightarrow g(Z_n) \xrightarrow{P} g(b)$
- $Z_n \xrightarrow{d} Z$, g continuous $\Rightarrow g(Z_n) \xrightarrow{d} g(Z)$

Slutsky: $X_n \xrightarrow{d} X$, $Y_n \xrightarrow{P} c$ (constant) $\Rightarrow X_nY_n \xrightarrow{d} cX$ and $X_n + Y_n \xrightarrow{d} X + c$.

Delta Method

If $\sqrt{n}(Z_n - \theta)/\sigma \xrightarrow{d} N(0,1)$ and $g'(\theta) \neq 0$:

$$\sqrt{n}\big(g(Z_n) - g(\theta)\big) \xrightarrow{d} N\big(0,\ \sigma^2[g'(\theta)]^2\big)$$

Requires $g'(\theta) \neq 0$. If $g'(\theta) = 0$, delta method breaks (need second-order).

Recipe: derivative of g, evaluate at θ, square it, multiply by original variance.

Method of Moments (MOM)

Set population moments = sample moments, solve:

1. $\mu_j(\theta) = E(X^j)$ for $j = 1, \ldots, k$ parameters
2. $\hat{\mu}_j = \frac{1}{n}\sum X_i^j$
3. Solve $\mu_j(\theta) = \hat{\mu}_j$

Dist	MOM Estimator
Ber(θ), Poi(λ)	$\hat{\theta} = \bar{X}$
$N(\mu, \sigma^2)$	$\hat{\mu} = \bar{X}$, $\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$
Exp(λ)	$\hat{\lambda} = 1/\bar{X}$
Gam(α, β)	use $\bar{X} = \alpha/\beta$, $\hat{\mu}_2 - \bar{X}^2 = \alpha/\beta^2$

MOM is consistent (LLN + CMT). Not unique. Not always optimal.

Maximum Likelihood Estimation (MLE)

$L(\theta) = \prod f(X_i, \theta)$, $\quad \ell(\theta) = \sum \log f(X_i, \theta)$

Set $\ell'(\theta) = 0$, verify $\ell''(\hat{\theta}) < 0$.

Dist	MLE
Ber(θ), Poi(θ)	$\hat{\theta} = \bar{X}$
$N(\mu, \sigma^2)$	$\hat{\mu} = \bar{X}$, $\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$ **(BIASED)**
Exp(θ) rate	$\hat{\theta} = 1/\bar{X}$
Unif$[0, \theta]$	$\hat{\theta} = X_{(n)} = \max(X_i)$ (by inspection)
Unif$[\theta, \theta{+}1]$	any $\theta \in [X_{(n)} - 1,\ X_{(1)}]$ (NOT unique)

Properties:

- **Invariance**: $g(\hat{\theta})$ is MLE of $g(\theta)$ for any function g
- **Consistent**: $\hat{\theta}_n \xrightarrow{P} \theta$ (under regularity conditions)
- **Caution**: MLE may NOT exist, NOT be unique, NOT be unbiased

When calculus fails: Unif$[0, \theta]$: $L(\theta) = \theta^{-n}\, I(\theta \geq X_{(n)})$ — decreasing in θ, so max at $\hat{\theta} = X_{(n)}$.

MSE & Comparing Estimators

$$\mathrm{MSE}(T_n) = \mathrm{Var}(T_n) + [\mathrm{bias}(T_n)]^2$$

For unbiased estimator: MSE = Var.

Estimator	Unbiased?	MVUE?	Notes
$\bar{X}$ for μ	Yes (always)	Yes (Normal)	Var $= \sigma^2/n$
$\frac{1}{n}\sum(X_i - \bar{X})^2$	No (for σ^2)	No	$E = \frac{n-1}{n}\sigma^2$
$s^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$	Yes (σ^2)	Yes (Normal)	Bessel's
$2\bar{X}$ for Unif$[0, \theta]$	Yes	No	MSE $= \theta^2/(3n)$
$X_{(n)}$ for Unif$[0, \theta]$	**No**	No	MSE $\ll 2\bar{X}$
$(1{+}1/n)X_{(n)}$	Yes	Yes (MVUE)	MSE $= \frac{\theta^2}{n(n+2)}$

Biased can beat unbiased (bias-variance tradeoff).

Unbiased estimators don't always exist: $\theta/(1-\theta)$ for Ber(θ) has none.

No universally best estimator in general.

Sufficient Statistics

T is sufficient for $\theta \;\Leftrightarrow\; f(\mathbf{x}, \theta) = u(\mathbf{x}) \cdot v(r(\mathbf{x}), \theta)$ **(Factorization Criterion)**

- $u(\mathbf{x})$: depends on data ONLY (not θ)
- $v(r(\mathbf{x}), \theta)$: θ enters ONLY through $r(\mathbf{x})$

Distribution	Sufficient Statistic
Poi(θ), Ber(θ), Exp(θ)	$\sum X_i$
$N(\mu, \sigma^2)$ both unknown	$(\sum X_i,\ \sum X_i^2)$
$N(\mu, \sigma^2$ known)	$\sum X_i$
Unif$[0, \theta]$	$X_{(n)} = \max(X_i)$
Gam(α known, β)	$\sum X_i$
Gam(α, β known)	$\prod X_i$

Rao-Blackwell Theorem

Given ANY estimator $\delta(X)$ and sufficient statistic T:

$$\delta_0(T) = E[\delta(X) \mid T]$$

Then: $\mathrm{MSE}(\delta_0) \leq \mathrm{MSE}(\delta)$ for all θ (never worse, often strictly better).

$\Rightarrow$ Always condition on sufficient statistics to improve estimators.

Bayesian Inference

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}: \quad \xi(\theta|x) \propto f(x|\theta) \cdot \xi(\theta)$$

Bayes est (squared error) $= E[\theta|X]$ (posterior mean).

Bayes est (absolute error) = posterior median.

Conjugate Priors

Likelihood	Conjugate Prior	Posterior
Ber(θ)	Beta(α, β)	Beta($\alpha{+}\sum x_i,\ \beta{+}n{-}\sum x_i$)
Poi(λ)	Gam(α, β)	Gam($\alpha{+}\sum x_i,\ \beta{+}n$)
$N(\mu, \sigma^2$ known)	$N(\mu_0, v_0^2)$	$N(\mu_n, v_n^2)$
Exp(θ)	Gam(α, β)	Gam($\alpha{+}n,\ \beta{+}\sum x_i$)

Normal–Normal Conjugate (σ^2 known)

Prior: $\mu \sim N(\mu_0, v_0^2)$. Data: $X_1, \ldots, X_n$ iid $N(\mu, \sigma^2)$.

Posterior: $\mu|X \sim N(\mu_n, v_n^2)$ where:

$$\mu_n = \frac{\mu_0/v_0^2 + n\bar{X}/\sigma^2}{1/v_0^2 + n/\sigma^2}, \qquad v_n^2 = \frac{1}{1/v_0^2 + n/\sigma^2}$$

- Posterior mean = weighted average of prior mean and $\bar{X}$, weighted by **precisions** (1/variance)
- As $n \to \infty$: posterior concentrates around $\bar{X}$ (data overwhelms prior)

- As $v_0^2 \to \infty$ (vague prior): posterior $\approx N(\bar{X}, \sigma^2/n)$

Sampling Distributions

Chi-Squared

$Z \sim N(0,1) \;\Rightarrow\; Z^2 \sim \chi_1^2$.
$Z_1, \ldots, Z_d$ iid $N(0,1) \;\Rightarrow\; \sum Z_i^2 \sim \chi_d^2$.
$\chi_d^2 = \text{Gam}(d/2,\ 1/2)$. $\quad E(\chi_d^2) = d, \quad \text{Var}(\chi_d^2) = 2d$.
Reproductive: $\sum \chi_{m_i}^2 \sim \chi_{\sum m_i}^2$ (independent).

Normal Sampling — CRITICAL RESULTS

$X_1, \ldots, X_n$ iid $N(\mu, \sigma^2)$:

1. $\bar{X} \sim N(\mu,\ \sigma^2/n)$
2. $s^2 \sim \dfrac{\sigma^2}{n-1}\chi_{n-1}^2$, equivalently $\dfrac{S^2}{\sigma^2} = \dfrac{\sum(X_i - \bar{X})^2}{\sigma^2} \sim \chi_{n-1}^2$
3. $\bar{X}$ and s^2 are **INDEPENDENT** (Proposition 6.3)

Independence of $\bar{X}$ and s^2 **characterizes** normality.

Theorem 6.4 (statement only): Z_i iid $N(0,1)$, A orthogonal $\Rightarrow$ $V = AZ$ has V_i iid $N(0,1)$ and $\sum V_i^2 = \sum Z_i^2$.

t-Distribution

$Z \sim N(0,1)$, $V \sim \chi_n^2$, $Z \perp V$:

$$T = \frac{Z}{\sqrt{V/n}} \sim t_n$$

Properties:

- Symmetric about 0 (T and $-T$ have same distribution)
- Heavier tails than $N(0,1)$ for finite n
- As $n \to \infty$: $t_n \to N(0,1)$ (by WLLN: $V/n \xrightarrow{P} 1$)

Theorem 6.5: X_i iid $N(\mu, \sigma^2)$:

$$\frac{\sqrt{n}(\bar{X} - \mu)}{s} \sim t_{n-1}$$

Confidence Intervals

(A, B) is level $1-\alpha$ CI if $P_\theta(A \le g(\theta) \le B) \ge 1-\alpha$ for all θ.

Method of Pivots

1. Construct $\Psi(\text{data}, \theta)$ with known distribution FREE of θ
2. Find quantiles: $P(q_{\beta_1} \le \Psi \le q_{1-\beta_2}) = 1-\alpha$ where $\beta_1 + \beta_2 = \alpha$
3. Invert the inequalities to isolate $g(\theta)$

Equal-tailed ($\beta_1 = \beta_2 = \alpha/2$) gives shortest CI for symmetric distributions.

Normal Mean CIs

Case	Pivot	Dist	CI
σ^2 known	$\frac{\sqrt{n}(\bar{X}-\mu)}{\sigma}$	$N(0,1)$	$\bar{X} \pm \frac{\sigma}{\sqrt{n}} z_{\alpha/2}$
σ^2 unknown	$\frac{\sqrt{n}(\bar{X}-\mu)}{s}$	t_{n-1}	$\bar{X} \pm \frac{s}{\sqrt{n}} t_{n-1,\alpha/2}$
Large n, any dist	$\frac{\sqrt{n}(\bar{X}-\mu)}{s}$	$\approx N(0,1)$	$\bar{X} \pm \frac{s}{\sqrt{n}} z_{\alpha/2}$

Other CIs

Bernoulli (large n): $\hat{\theta} \pm \sqrt{\dfrac{\hat{\theta}(1-\hat{\theta})}{n-1}}\, z_{\alpha/2}$

CI for σ^2 (normal data): Pivot $(n-1)s^2/\sigma^2 \sim \chi_{n-1}^2$.

$$\text{CI}: \quad \left(\frac{(n-1)s^2}{\chi_{n-1,\alpha/2}^2}, \quad \frac{(n-1)s^2}{\chi_{n-1,1-\alpha/2}^2} \right)$$

CI Properties

- Width (normal, σ known): $2\sigma\, z_{\alpha/2}/\sqrt{n}$
- Higher confidence (smaller α) $\to$ WIDER CI
- Larger $n \to$ NARROWER CI (width $\propto 1/\sqrt{n}$)
- $4\times$ sample size $\Rightarrow$ half the CI width
- $\alpha = 0 \Rightarrow z_{\alpha/2} = \infty \Rightarrow$ infinitely wide

CI Interpretation (Classic T/F Trap!)

CORRECT: "We have *confidence* γ that θ lies in (a, b)."

WRONG: "θ lies in (a, b) with *probability* γ." — θ is FIXED (not random), the interval is random. After observing data, it either contains θ or it doesn't.

Cramér-Rao Inequality

Score: $\dot{\ell}(x, \theta) = \frac{\partial}{\partial\theta} \log f(x, \theta)$. $\quad E[\dot{\ell}] = 0$. $\quad$ Fisher info: $I(\theta) = E[\dot{\ell}^2]$.
For n iid observations: total info $= n\, I(\theta)$.

Regularity Conditions (KNOW THESE)

(A.1) Support $\{x : f(x, \theta) > 0\}$ does NOT depend on θ.
Excludes: Unif$[0, \theta]$, Unif$[\theta, \theta{+}1]$ — support changes with θ.

(A.2) Can interchange $\frac{\partial}{\partial\theta}$ and $\int$:

$$\frac{\partial}{\partial\theta} \int W(\mathbf{x})\, f(\mathbf{x}, \theta)\, d\mathbf{x} = \int W(\mathbf{x})\, \frac{\partial f}{\partial\theta}(\mathbf{x}, \theta)\, d\mathbf{x}$$

(A.3) $\frac{\partial}{\partial\theta} \log f(x, \theta)$ exists for all $x \in A$ and all $\theta \in \Omega$, as a finite quantity.

The CRLB

If $T(\mathbf{X}_n)$ is UNBIASED for $g(\theta)$, and A.1–A.3 hold, and $I(\theta) < \infty$:

$$\boxed{\text{Var}_\theta(T) \;\ge\; \frac{[g'(\theta)]^2}{n\, I(\theta)}}$$

Special case $g(\theta) = \theta$: $\quad \text{Var}(T) \ge 1/(n\, I(\theta))$.
Achieves the bound $\Rightarrow$ **MVUE** (efficient estimator).

Common Fisher Information

Dist	$f(x, \theta)$	$\dot{\ell}(x, \theta)$	$I(\theta)$	CRLB
Ber(θ)	$\theta^x(1-\theta)^{1-x}$	$\frac{x-\theta}{\theta(1-\theta)}$	$\frac{1}{\theta(1-\theta)}$	$\frac{\theta(1-\theta)}{n}$
Poi(θ)	$e^{-\theta}\theta^x/x!$	$x/\theta - 1$	$1/\theta$	θ/n
$N(\mu, \sigma^2)$	normal pdf	$(x-\mu)/\sigma^2$	$1/\sigma^2$	σ^2/n
Exp(θ)	$\theta e^{-\theta x}$	$1/\theta - x$	$1/\theta^2$	θ^2/n

$\bar{X}$ is MVUE for: μ in Normal, θ in Bernoulli, θ in Poisson — all achieve CRLB.

True/False Traps

1. **F**: "Conv in dist $\Rightarrow$ conv in prob" (only true if limit is a constant)
2. **F**: "MLE is always unbiased" ($\hat{\sigma}^2$ is biased; $X_{(n)}$ is biased)
3. **F**: "MLE always exists and is unique" (Unif$[\theta, \theta{+}1]$: not unique)
4. **F**: "Unbiased is always better than biased" (biased $X_{(n)}$ beats $2\bar{X}$ for Unif)

5. **F**: "$\theta \in$ CI with prob $1-\alpha$" (θ is fixed; say "confidence")
6. **T**: MLE invariance: $g(\hat{\theta})$ is MLE of $g(\theta)$
7. **T**: MLE is consistent under regularity conditions
8. **T**: Rao-Blackwell improves estimators via sufficiency
9. **F**: "Unbiased estimators always exist" ($\theta/(1-\theta)$ for Ber: none)
10. **F**: "CRLB applies to all distributions" (fails for Uniform — A.1 violated)
11. **T**: $\bar{X} \perp s^2$ for normal data (Prop 6.3; characterizes normality)
12. **F**: "CLT says $\bar{X}$ is exactly normal" (approx only; exact only if X_i are normal)
13. **T**: Sufficient stat can be vector-valued (N: $(\sum X_i, \sum X_i^2)$)
14. **F**: "Consistent $\Rightarrow$ unbiased" (can be biased for each finite n)
15. **T**: As $n \to \infty$, posterior $\to$ true θ regardless of prior

Computation Recipes

1. **Find MLE**: Write $L(\theta)$, take log, $\ell'(\theta) = 0$, check $\ell'' < 0$. Uniform: by inspection.
2. **Find MOM**: Set $E(X^j) = \frac{1}{n}\sum X_i^j$, solve for parameters.
3. **Unbiased?**: Compute $E(\text{est})$, check if $= g(\theta)$ for ALL θ.
4. **Sufficient stat**: Write joint pdf, factor as $u(\mathbf{x}) \cdot v(r(\mathbf{x}), \theta)$.
5. **Compare estimators**: MSE = Var + Bias^2 for each, compare across all θ.
6. **Show consistency**: Show $\hat{\theta}_n \xrightarrow{P} \theta$ via LLN, Chebyshev, or CMT.
7. **Asymptotic dist**: CLT for $\bar{X}$, then delta method if transformed.
8. **Construct CI**: Pick pivot (Z, t, or χ^2), find quantiles, invert.
9. **Fisher information**: $\dot{\ell} = \partial/\partial\theta \log f$, then $I(\theta) = E[\dot{\ell}^2]$.
10. **Is it MVUE?**: Check unbiased + Var = CRLB, or use Rao-Blackwell.
11. **Find posterior**: Write $\propto$ likelihood $\times$ prior, identify the distribution kernel.
12. **Bayes estimator**: Sq. error = posterior mean; abs. error = posterior median.
13. **Interpret CI**: Say "confidence," NOT "probability." θ is fixed.
14. **Rao-Blackwell**: Find sufficient T, compute $E[\delta(X)|T]$.
15. **Sample size**: Use Chebyshev or CI width formula. $n \geq \sigma^2/(\varepsilon^2\delta)$.